INCH-POUND

MIL-DTL-17/183C
w/AMENDMENT 1
22 June 2019
SUPERSEDING
MIL-DTL-17/183C
25 July 2014

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/183-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

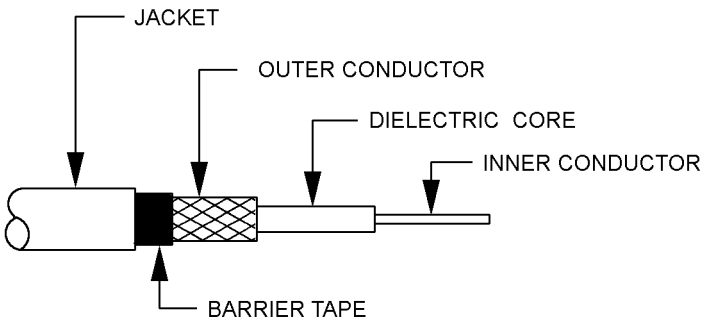


FIGURE 1. General configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.

TABLE I. Description.

Component	Construction
Inner conductor	Nineteen strands of tinned copper wire, each strand .0072 inch diameter. Overall diameter: .0355 inch ±0.0020
Dielectric core	Type A-1: Solid, polyethylene. Diameter: .116 inch ±0.004
Outer conductor	Single braid of AWG No. 36, tinned copper wire Diameter: .145 inch maximum Coverage: 92.8% nominal Carriers: 12 Ends: 9 Picks/inch: 7.7 ±10% Alternate Coverage: 94.2% nominal Carriers: 16 Ends: 7 Picks/inch: 10.3 ±10%
Barrier tape	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape shall be applied with a 50% lap, aluminum face toward the outer conductor. Diameter: .155 inch maximum
Jacket	Type XIV, Cross-linked polyolefin Diameter: .195 inch ±0.004

ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 1GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -30°C to +85°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 1.48 ohms per 100 feet.

Elongation: 15 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in general purpose low temperature applications. (See connector series “TNC”, “BNC” and “SMA” in accordance with MIL-PRF-39012.)

REQUIREMENTS (See table II)

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.

Visual and mechanical examination: Applicable.

Diameter: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 5 pounds, minimum; 15 pounds, maximum.

Electrical and mechanical:

Continuity: Applicable.

Spark test: 5,000 V rms, minimum.

Voltage withstanding: 5,000 V rms, minimum.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

RF transmission loss (attenuation): See figure 2.

Standing wave ratio (return loss): See figure 3.

Capacitance: 32.2 pF per foot.

Capacitance stability:	Not applicable.
Capacitance unbalance:	Not applicable.
Transmission unbalance:	Not applicable.
Inductance:	Not applicable
Mechanically induced noise voltage:	Not applicable.
Time delay:	Not applicable.

Aging stability: +98ºC ± 2ºC.

Stress crack resistance:	Not applicable.
Outer conductor integrity:	Not applicable.

Cold bend: -30°C ± 2°C.

Dimensional stability: +85ºC ± 2ºC.

Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.

Contamination:	Not applicable.
Bendability:	Not applicable.
Flammability:	Not applicable.

Flame propagation: Applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.

Acid gas generation: 2 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50.

Elongation, percent of unaged minimum: 50.

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness: (Type A) 80 minimum.

Weathering: Applicable.

Abrasion resistance: 75 cycles minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation: 160 percent, minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60.

Elongation, percent minimum: 60.

Hot oil immersion:

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent minimum.

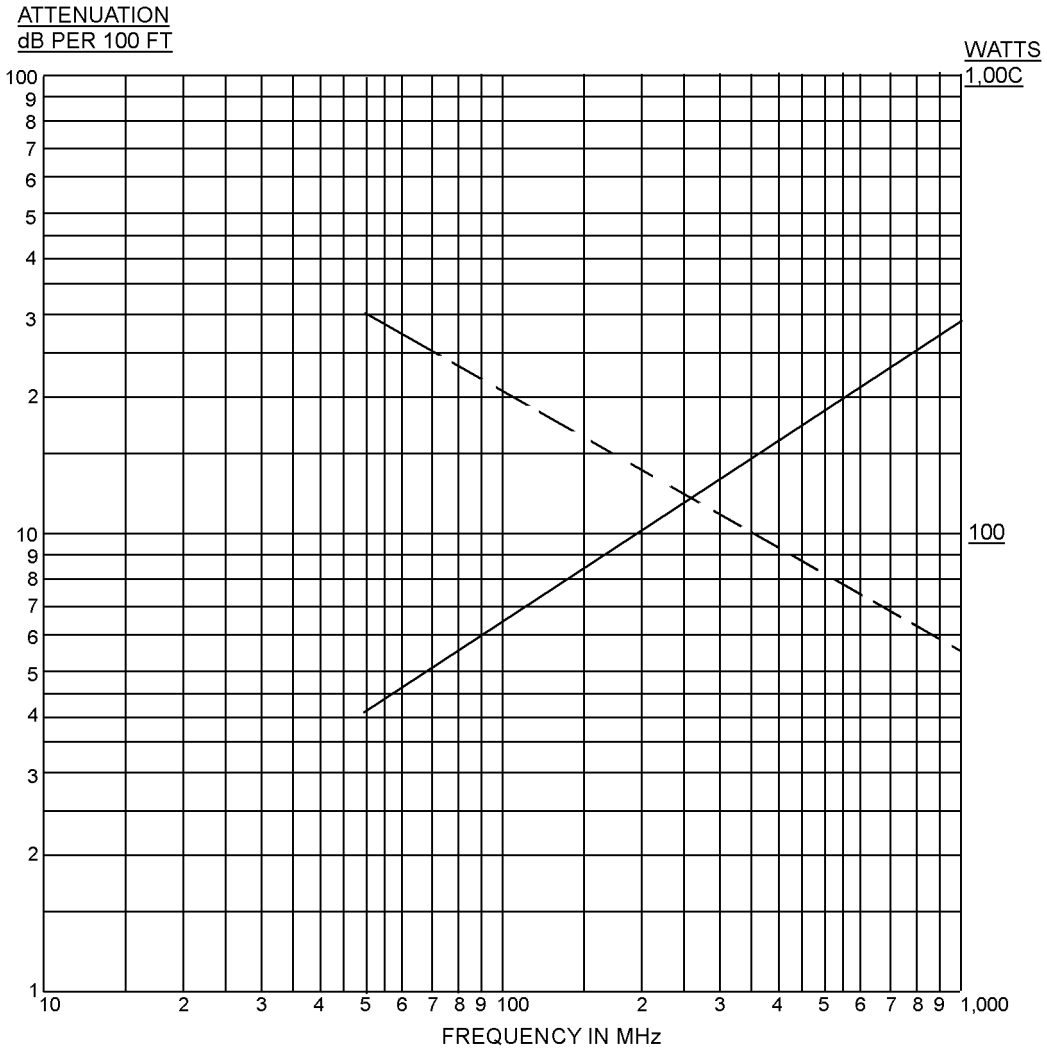
Weight: 3 pounds per 100 feet, maximum.

Marking: Applicable.

Workmanship: Applicable.

Part or Identifying Number (PIN): M17/183-00001.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.



MAXIMUM POWER - - - - - - - - - - AT SEA LEVEL AND 25° C

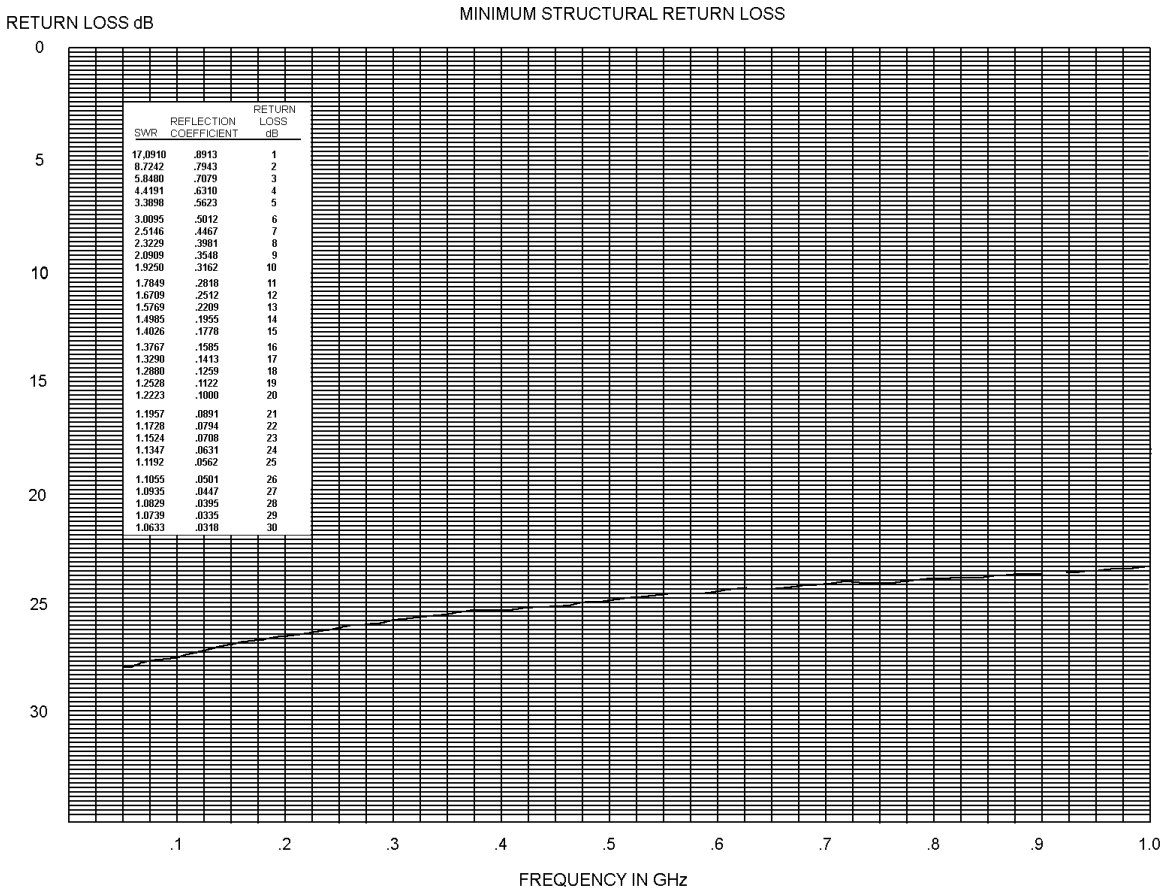
MAXIMUM ATTENUATION _____

Tabulated values are for reference only
The values on the chart represent the requirements.

Frequency MHz	Attenuation db	Power Watts
50	4.0	300
100	6.5	200
400	17.0	90
1000	28.0	33

FIGURE 2. Power rating at sea level and 25ºC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.



SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.0910	.8913	1
8.7242	.7943	2
5.8480	.7079	3
4.4191	.6310	4
3.3898	.5623	5
3.0095	.5012	6
2.5146	.4467	7
2.3229	.3981	8
2.0909	.3548	9
1.9250	.3162	10
1.7849	.2818	11
1.6709	.2512	12
1.5769	.2209	13
1.4985	.1955	14
1.4026	.1778	15
1.3767	.1585	16
1.3290	.1413	17
1.2880	.1259	18
1.2528	.1122	19
1.2223	.1000	20
1.1957	.0891	21
1.1728	.0794	22
1.1524	.0708	23
1.1347	.0631	24
1.1192	.0562	25
1.1055	.0501	26
1.0935	.0447	27
1.0829	.0395	28
1.0739	.0335	29
1.0633	.0318	30

FIGURE 3. Structural return loss.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.

TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.

TABLE II. Qualification and Group A, B, and C Table - Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Time delay	N/A				
	Aging stability	3.7.15	4.8.16	4	Per Table VIII	C
	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	3.7.23	4.8.25	1		
	Acid gas generation	3.7.24	4.8.26	1		
	Halogen content	3.7.25	4.8.27	1		
	Immersion tests	3.7.26	4.8.28	1		
	Smoke index	3.7.27	4.8.29	1		
	Toxicity Index	3.7.28	4.8.30	1		
	Durometer hardness	3.7.29	4.8.31	1		
	Weathering	3.7.30	4.8.32	1		
	Abrasion resistance	3.7.31	4.8.33	5		
	Tear strength	3.7.32	4.8.34	6	Per Table VIII	B
	Heat distortion	3.7.33	4.8.35	1	Per Table VIII	C
	Physicals (aged)	3.7.34	4.8.36	1	Per Table VIII	C
	Tensile strength and elongation	3.7.35	4.8.37	1	Per Table VIII	C
	Weight	3.9	4.8.24	1	Per Table VIII	B

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.

Amendment notations: The margins of this specification are marked with vertical lines to indicate where modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity
DLA - CC

(Project 6145-2019-020)

Review activities:
Army - AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:26Z
Check the source to verify that this is the current version before use.